DEVELOPMENT

BobaLink — Development Guide

Revision: 1 **Last modified:** 2026-06-10T21:00:00Z **Scope:** Architecture, layout, testing, and contribution rules for BobaLink (`extension/`). **Authority:** `extension/package.json`, `extension/wxt.config.ts`, `docs/browser_extension/IMPLEMENTATION_PLAN.md`, `docs/browser_extension/Status.md` (Rev 4), project `CLAUDE.md` + `constitution/`.

> Accuracy note (§11.4.6): the architecture and commands below reflect the real tree at HEAD `2011810`. Phases still open are marked **(in progress)**.

Stack

- **WXT** (`wxt ^0.19.0`) — the WebExtension build framework; generates the MV3 manifest from `wxt.config.ts` and bundles entrypoints into `.output/`.
- **TypeScript** (`^5.7.0`, strict) — `npm run compile` is `tsc --noEmit`.
- **Vitest** (`^2.1.8`) + `@vitest/coverage-v8` — unit/integration/etc. tests.
- **Playwright** (`@playwright/test ^1.49.0`) — end-to-end (real MV3 extension load).
- **ESLint** (`^9.17.0`) + Prettier — lint/format.
- Runtime dependency: `webextension-polyfill`.
- Node ≥ 20, ESM (`"type": "module"`).

Architecture

The pipeline, end to end:

```
page DOM
  → parser        (magnet / bencode / .torrent → SHA-1 infohash)
```

```
  → scanner       (link-scanner + text-scanner, MutationObserver,
site selectors)
  → orchestrator  (cross-scanner dedup by infohash)
  → background    (service-worker message router; context menus;
commands; alarms)
  → boba-client   (POST to the Boba merge service on :7187; token
header; retry)
  → queue         (offline FIFO; persists across SW restart; auth-
injected send)
```

Module map (under `extension/src/`):

- **parsers** — magnet URI parsing, bencode decoding, `.torrent` → SHA-1 infohash (`SHA-1(bencode(info))` via WebCrypto). Dedup key is the lowercase infohash.
- **scanners** — `link-scanner` / `text-scanner` + `content/scanner`; stable infohash-based IDs; Shadow-DOM/iframe aware; 500 ms MutationObserver debounce.
- **orchestrator** — runs the scanners and deduplicates across them.
- **`background/index.ts`** — the service-worker message router: dispatches context-menu clicks (`bobalink-send` / `bobalink-send-all` / `bobalink-send-group`), keyboard commands (`send-to-boba` / `scan-page` / `open-dashboard`), and `chrome.alarms` keep-alive + periodic health checks.
- **`api/boba-client.ts`** — the Boba merge-service client (Phase 4, **in progress**): posts magnets/URLs to `:7187`, presents the optional decrypted Boba API token as an `Authorization`/`X-Boba-Token` header, with retry/backoff and an `AbortController` timeout.
- **`api/queue.ts`** — the offline FIFO queue; enqueues on send failure and flushes when the backend recovers; durable across service-worker restart.
- **`tabgroups/index.ts`** — dedupe + batch dispatch across a Chrome tab group, invoked from the background `MENU_SEND_GROUP` handler.
- **`shared/`** — `constants.ts`, `crypto.ts` (AES-GCM-256 / PBKDF2), `errors.ts`, `events.ts`, `logger.ts`, `storage.ts`, `utils.ts`.
- **`options/options.ts`** + **`popup/`** — the 7-tab options page and the detected-list popup with Send / Send-All.

WXT entrypoints (thin wrappers)

WXT discovers entrypoints under `extension/src/entrypoints/`:

```
src/entrypoints/
  background.ts          → background.js service worker
  content.ts             → content-scripts/content.js
  popup/index.html       → popup.html
  options/index.html     → options.html
```

These are **thin wrappers** over the logic modules listed above — the heavy logic lives in `src/{parsers,scanners,background,api,options,popup,shared,...}/`, so the existing unit tests import the logic modules directly and stay valid regardless of WXT's entrypoint plumbing (`Status.md` Rev 4, "WXT build wiring — COMPLETE").

`wxt.config.ts` sets `srcDir: "src"`, `outDir: ".output"`, the MV3 manifest (least-privilege permissions, localhost-only `host_permissions`, the CSP, the three keyboard `commands`), and `@wxt-dev/auto-icons` to rasterize icon sizes.

Running tests

From `extension/`:

```
npx vitest run          # the full unit/integration/security/perf/etc. suite
npm run test            # === vitest run --coverage
npm run test:watch      # vitest in watch mode
npm run test:live       # vitest run --config tests/live/vitest.live.config.ts
npm run test:e2e        # playwright test (real MV3 extension load)
```

- The Vitest corpus lives under `extension/tests/{unit,perf,stress,chaos,integration,security,e2e}` and alongside source in `src/**`. As of `Status.md` Rev 4 a same-session `npx vitest run` reported `Tests 413 passed (413)` across 37 spec files.
- **`test:live`** targets a live backend; live integration uses a `require_backend(7187)` guard that **SKIPs with reason** when `:7187` is down (never fail-open, §11.4.3/§11.4.68).
- **`test:e2e`** loads the real built extension via Playwright (`launchPersistentContext` + `--load-extension`, real id from the SW target). The MV3-load e2e is **operator-gated SKIP** in a headless sandbox where extension load is unsupported.
- Anti-bluff is the gate: each test asserts a user-observable outcome and must fail against a no-op stub (`IMPLEMENTATION_PLAN.md` §0.2). Challenges under `challenges/extension/` and `challenges/security/` drive the real orchestrator/client/audit end-to-end and are mutation-verified.

Type + lint, run by `ci-ext.sh` and individually:

```
npm run compile         # tsc --noEmit
npm run lint            # eslint src/ tests/ --ext .ts
npm run format:check    # prettier --check "src/**/*.ts"
```

Constitution constraints (read before contributing)

BobaLink inherits the Boba `CLAUDE.md` and the `constitution/` submodule. The load-bearing rules for this subproject:

- **Anti-bluff (§11.4).** Every test/Challenge must prove the feature works for the end user — assert user-observable outcomes (DB rows, response bodies, DOM text), and each test must FAIL against a no-op stub. The reference suite's 10 known bluffs are remediated, not adopted.
- **TDD (§11.4.43/§11.4.115).** Write the failing RED test first (reproducing the real defect / driving the real feature), watch it fail, then write minimal code to GREEN. "Test added after the fix" is a bluff.
- **NO CI/CD — ever (Hard Stop §1).** No `.github/workflows/`, no pipelines, no git hooks. Validation is the manual `extension/ci-ext.sh`.
- **No force-push (§11.4.113); fetch-first (§11.4.71); SSH remotes only.**
- **§11.4.10 credentials.** Delegate-by-default; no embedded/decryptable secret by default; no fixed/empty-passphrase decrypt; no real credential in any test/fixture/doc.
- **100% multi-type coverage (§11.4.27/§11.4.85).** unit/integration/e2e/security/ load/scaling/chaos/stress/performance/benchmark/UI/UX + Challenges + HelixQA.
- **Mandatory code review (§11.4.125/§11.4.134/§11.4.142)** of every change, iterate-until-GO, before it is "complete".

The single real backend is the **Boba merge service on `:7187`** — never qBittorrent directly, never port 8080 (the reference `:8080` values were all retargeted to `:7187`; see `src/shared/constants.ts`).

Where the docs / Status / ledger live

All under `docs/browser_extension/`:

- **`Status.md`** (+ `.html`/`.pdf`) — the authoritative per-phase status (Rev 4 is the real state).
- **`Status_Summary.md`** (+ siblings) — the two-audience summary companion.
- **`IMPLEMENTATION_PLAN.md`** (+ siblings) — the 9-phase master plan with the hard constraints, backend additions (BE-1/2/3), and adopt/refactor/rewrite map.
- **`coverage_ledger.md`** (+ siblings) — the feature × test-type coverage ledger.
- **`USER_GUIDE.md`**, **`INSTALL.md`**, **`DEVELOPMENT.md`** (this file) — the Phase-9 user/install/dev docs (+ `.html`/`.pdf` siblings per §11.4.65).

- Deeper provenance: `docs/browser_extension/_analysis/` and `_plan/`.

The script companion for the gate is `docs/scripts/ci-ext.md` (§11.4.18). HelixQA banks: `submodules/helixqa/banks/boba-bobalink.yaml` (symlinked into `challenges/helixqa-banks/`).